

Frame structure for the wedge Stolarsky energy

ETF minimizers and a first non-ETF obstruction in arXiv:1511.08452

Automated Mathematics Run `fa_banach_001`, lane 2

19 July 2026

Status. Candidate partial solution, likely valid, subject to human review. It characterizes every minimizer for $N \leq d + 2$ and every parameter pair admitting a real equiangular tight frame. At the first non-ETF pair, $(N, d) = (5, 2)$, it also proves that no global minimizer is a tight frame.

1 Source and exact question

The source is Bilyk and Lacey, *One Bit Sensing, Discrepancy, and Stolarsky Principle* [1]. For $Z = \{z_1, \dots, z_N\} \subset \mathbb{S}^d$ and normalized geodesic distance

$$d(x, y) = \frac{1}{\pi} \arccos \langle x, y \rangle,$$

Theorem 1.23 identifies minimization of the L^2 wedge discrepancy with minimization of

$$\mathcal{E}(Z) = \sum_{i,j=1}^N \left(\frac{1}{2} - d(z_i, z_j) \right)^2. \quad (1)$$

The harmless source normalization N^{-2} is omitted. Immediately afterward, the authors ask whether the minimizers admit a geometric or functional-analytic characterization and whether known distributions give good energy values.

2 Proof intuition

The source potential is a nonlinear frame potential in disguise. The identity $\frac{1}{2} - \pi^{-1} \arccos t = \pi^{-1} \arcsin t$ shows that it applies a single strictly convex increasing function to every squared correlation $|\langle z_i, z_j \rangle|^2$. The classical frame-potential bound controls the mean squared correlation; Jensen then says that the nonlinear energy is smallest precisely when all off-diagonal squared correlations are equal. Equality in the frame bound simultaneously forces tightness. These two equality conditions are exactly equiangular tightness.

When no ETF exists, however, tightness need not survive. For five vectors in $\mathbb{R}^3$, Naimark complementation turns every tight frame into five points on a circle with a completely monotone pair potential. Universal optimality of the regular pentagon therefore gives the least energy possible among all tight frames. A one-parameter cyclic deformation has positive energy derivative at that pentagonal tight frame, so moving away from tightness strictly lowers the energy. The general non-ETF classification remains open.

Theorem 1.23 (Stolarsky principle for wedges). *For any finite set $Z = \{z_1, \dots, z_N\} \subset \mathbb{S}^d$, the following relation holds*

$$(1.24) \quad \|\Delta_Z(x, y)\|_2^2 = \frac{1}{N^2} \sum_{i,j=1}^N \left(\frac{1}{2} - d(z_i, z_j) \right)^2 - \int_{\mathbb{S}^d} \int_{\mathbb{S}^d} \left(\frac{1}{2} - d(x, y) \right)^2 d\sigma(x) d\sigma(y).$$

Minimizing the L^2 average of the wedge discrepancy associated to the tessellation of the sphere is thus equivalent to minimizing the discrete potential energy of Z induced by the potential $P(x, y) = \left(\frac{1}{2} - d(x, y) \right)^2$. Intuitively, one would like to make the elements of Z ‘as orthogonal as possible’ on the average.

First of all, this suggests natural candidates for tessellations that are good or optimal on the average, e.g., spherical codes (sets $X \subset \mathbb{S}^d$ such that all $x, y \in X$ satisfy $x \cdot y < \mu$ for some parameter $\mu \leq 1$), see [37, Chapter 5] and references therein, or equiangular lines (sets $X \subset \mathbb{S}^d$ such that all $x, y \in X$ satisfy $|x \cdot y| = \mu$ for some fixed $\mu \in [0, 1)$), see [17].

This also brings up connections to frame theory. Benedetto and Fickus [8] proved that a set $Z = \{z_1, \dots, z_N\} \subset \mathbb{S}^d$ forms a normalized tight frame (i.e. there exists a constant $A > 0$ such that for every $x \in \mathbb{R}^{d+1}$ an analog of Parseval’s identity holds: $A\|x\|^2 = \sum_{i=1}^N |x \cdot z_i|^2$) if and only if Z is a minimizer of a discrete energy known as the *total frame potential*:

$$TP(Z) = \sum_{i,j=1}^N |z_i \cdot z_j|^2,$$

which looks unmistakably similar to the discrete energy on the right-hand side of (1.24).

It is not known yet whether the minimizers of (1.24) admit a similar geometric or functional-analytic characterization, or if some of the known distributions yield reasonable values for this energy. These are interesting questions to be addressed in future research. We prove Theorem 1.23 in §3.

Figure 1: The wedge energy and complete future-work question on source PDF page 9 [1].

3 Sharp ETF bound and equality characterization

Put $m = d + 1$ and define

$$\Phi(s) = (\arcsin \sqrt{s})^2, \quad 0 \leq s \leq 1.$$

Then

$$\mathcal{E}(Z) = \frac{1}{\pi^2} \sum_{i,j=1}^N \Phi(|\langle z_i, z_j \rangle|^2). \quad (2)$$

Theorem 1. *Let $z_1, \dots, z_N$ be unit vectors in $\mathbb{R}^m$.*

If $N \leq m$, then

$$\mathcal{E}(Z) \geq \frac{N}{4},$$

with equality if and only if the vectors are pairwise orthogonal.

If $N > m$, then

$$\mathcal{E}(Z) \geq \frac{N}{4} + \frac{N(N-1)}{\pi^2} \arcsin^2 \sqrt{\frac{N-m}{m(N-1)}}. \quad (3)$$

Equality holds if and only if Z is a real unit-norm equiangular tight frame:

$$\sum_{i=1}^N z_i z_i^\top = \frac{N}{m} I_m, \quad |\langle z_i, z_j \rangle|^2 = \frac{N-m}{m(N-1)} \quad (i \neq j). \quad (4)$$

Consequently, whenever such an ETF exists, the ETFs are exactly the global minimizers of the wedge energy.

Proof. First we verify the needed convexity. Write $s = \sin^2 u$ with $0 < u < \pi/2$. Direct differentiation gives

$$\Phi'(s) = \frac{2u}{\sin 2u}.$$

The derivative of the right-hand side with respect to u has the sign of

$$q(u) = \sin 2u - 2u \cos 2u.$$

Since $q(0) = 0$ and $q'(u) = 4u \sin 2u > 0$, the function Φ is strictly increasing and strictly convex on $[0, 1]$ (with the endpoint statements by continuity).

The diagonal terms in (2) contribute $N\Phi(1)/\pi^2 = N/4$. Thus for $N \leq m$ the asserted lower bound is immediate, and equality occurs exactly when every off-diagonal inner product vanishes.

Now suppose $N > m$. Let

$$S = \sum_{i=1}^N z_i z_i^\top.$$

Since S is positive semidefinite and $\text{tr } S = N$, Cauchy–Schwarz on its m eigenvalues gives the frame-potential bound

$$\sum_{i,j=1}^N |\langle z_i, z_j \rangle|^2 = \text{tr}(S^2) \geq \frac{N^2}{m}. \quad (5)$$

Equality holds exactly when $S = (N/m)I_m$. Removing the N diagonal terms,

$$\frac{1}{N(N-1)} \sum_{i \neq j} |\langle z_i, z_j \rangle|^2 \geq \frac{N-m}{m(N-1)}.$$

Jensen's inequality and strict monotonicity of Φ now yield

$$\begin{aligned} \sum_{i \neq j} \Phi(|\langle z_i, z_j \rangle|^2) &\geq N(N-1) \Phi \left(\frac{1}{N(N-1)} \sum_{i \neq j} |\langle z_i, z_j \rangle|^2 \right) \\ &\geq N(N-1) \Phi \left(\frac{N-m}{m(N-1)} \right). \end{aligned}$$

Together with the diagonal contribution this is (3).

Because Φ is strictly convex, equality in Jensen occurs exactly when all off-diagonal squared inner products are equal. Equality in the second line also requires equality in (5), hence tightness. These conditions are precisely (4), and an ETF plainly attains every equality above. $\square$

Corollary 2 (complete low-cardinality classification). *For the source problem on $\mathbb{S}^d$:*

1. if $N \leq d+1$, the minimizers are exactly orthonormal N -tuples;

2. if $N = d + 2$, the minimizers are exactly regular simplices, up to an orthogonal transformation and independent sign changes of the vectors.

Proof. Only the second claim needs comment. For $m = d + 1$ and $N = m + 1$, a regular simplex is an ETF, so Theorem 1 applies. Conversely, the Gram matrix G of an equality case has eigenvalues $(m + 1)/m$ with multiplicity m and 0 with multiplicity one. If u is its unit null vector, then

$$G = \frac{m + 1}{m}(I - uu^\top).$$

The unit diagonal forces $|u_i| = 1/\sqrt{m + 1}$. Independent sign changes make all u_i positive, after which every off-diagonal entry of G is $-1/m$. This is the regular-simplex Gram matrix; equality of Gram matrices gives an orthogonal equivalence. $\square$

4 The first non-ETF case: global minimizers are not tight

The lower bound in Theorem 1 cannot be attained for five vectors in $\mathbb{R}^3$. Indeed, a real $(5, 3)$ ETF would have a Naimark complement giving five equiangular lines in $\mathbb{R}^2$, whereas at most three such lines exist. The next theorem shows that replacing equiangularity by tightness is not the right generalization.

Theorem 3 (non-tightness obstruction). *For $N = 5$ on $\mathbb{S}^2$, no global minimizer of the wedge energy is a unit-norm tight frame.*

More precisely, among all unit-norm tight frames of five vectors in $\mathbb{R}^3$, the minimum energy is

$$E_{\text{tight}} = \frac{5}{4} + \frac{10}{\pi^2} \left[\arcsin^2\left(\frac{1 + \sqrt{5}}{6}\right) + \arcsin^2\left(\frac{\sqrt{5} - 1}{6}\right) \right] = 1.622422282596 \dots \quad (6)$$

It is attained by the pentagonal harmonic tight frame

$$z_k(q) = \left(\sqrt{1 - q} \cos \frac{2\pi k}{5}, \sqrt{1 - q} \sin \frac{2\pi k}{5}, \sqrt{q} \right), \quad 0 \leq k < 5, \quad (7)$$

at $q = 1/3$. There are values $q < 1/3$ arbitrarily close to $1/3$ for which $\mathcal{E}(\{z_k(q)\}) < E_{\text{tight}}$.

Proof. We first minimize over tight frames. Let G be the Gram matrix of an arbitrary unit-norm tight frame of five vectors in $\mathbb{R}^3$. Then $P = (3/5)G$ is an orthogonal projection of rank three with diagonal $3/5$. Its complementary projection $Q = I - P$ has rank two and diagonal $2/5$. Consequently there are unit vectors $y_1, \dots, y_5 \in \mathbb{S}^1$ whose Gram matrix H satisfies

$$H = \frac{5}{2}Q, \quad H_{ij} = -\frac{3}{2}G_{ij} \quad (i \neq j).$$

This is the usual Naimark complement construction.

Write $y_i = (\cos \theta_i, \sin \theta_i)$ and double the angles: $x_i = (\cos 2\theta_i, \sin 2\theta_i) \in \mathbb{S}^1$. If $r_{ij} = \|x_i - x_j\|^2$, then for $i \neq j$

$$|G_{ij}|^2 = \frac{4}{9}|H_{ij}|^2 = \frac{4}{9} \cos^2(\theta_i - \theta_j) = \frac{4}{9} - \frac{1}{9}r_{ij}. \quad (8)$$

The power series

$$\Phi(s) = (\arcsin \sqrt{s})^2 = \frac{1}{2} \sum_{n=1}^{\infty} \frac{4^n}{n^2 \binom{2n}{n}} s^n \quad (9)$$

has strictly positive coefficients. Hence $r \mapsto \Phi(4/9 - r/9)$ is strictly completely monotone on $[0, 4]$. The regular pentagon is a two-distance spherical 4-design on $\mathbb{S}^1$, so the sharp-configuration theorem of Cohn and Kumar [4] says that it minimizes the sum of this potential over every five-point configuration on $\mathbb{S}^1$. Applying that result to the doubled-angle points in (8) proves the lower bound (6). Direct summation in (7) at $q = 1/3$ gives equality: five unordered pairs have correlation magnitude $(1 + \sqrt{5})/6$, and the other five have magnitude $(\sqrt{5} - 1)/6$; its frame operator is $(5/3)I_3$.

It remains to exhibit the descent away from tightness. Put

$$c = \cos \frac{2\pi}{5} = \frac{\sqrt{5} - 1}{4}, \quad C = \cos \frac{4\pi}{5} = -\frac{\sqrt{5} + 1}{4}.$$

Near $q = 1/3$, the two correlation magnitudes of (7) are

$$a(q) = q + (1 - q)c, \quad b(q) = -q - (1 - q)C,$$

each occurring for five unordered pairs. Thus

$$E(q) = \frac{5}{4} + \frac{10}{\pi^2} (\arcsin^2 a(q) + \arcsin^2 b(q)). \quad (10)$$

Let $A(x) = \arcsin(x)/\sqrt{1 - x^2}$. At $q = 1/3$,

$$a = \frac{1 + \sqrt{5}}{6}, \quad b = \frac{\sqrt{5} - 1}{6},$$

and differentiation gives

$$E'(1/3) = \frac{20}{\pi^2} \left[\frac{5 - \sqrt{5}}{4} A(a) - \frac{5 + \sqrt{5}}{4} A(b) \right] > 0. \quad (11)$$

For completeness, the sign is elementary. On $(0, 1)$, $\arcsin x > x + x^3/6$ by its positive Taylor series, while $\arcsin x < x/\sqrt{1 - x^2}$ by integration. After applying these bounds to the first and second terms, respectively, the square of the lower bound on the positive term minus the square of the upper bound on the negative term is

$$\frac{115}{10692} + \frac{3415\sqrt{5}}{128304} > 0.$$

Both bounds are positive, proving (11). Hence $E(1/3 - \varepsilon) < E(1/3) = E_{\text{tight}}$ for all sufficiently small positive ε . Since every tight frame has energy at least E_{tight} , no global minimizer can be tight. $\square$

Corollary 4 (exact cyclic optimization). *On the interval where $a(q), b(q) > 0$, the cyclic energy (10) is strictly convex and has a unique minimizer $q_* = 0.314800327343\dots$. It is equivalently characterized by*

$$(1 - c)A(a(q_*)) = (1 - C)A(b(q_*)).$$

At this point the two correlation magnitudes are $0.526538670734\dots$ and $0.239537852377\dots$, and $E(q_) = 1.620843159568\dots$*

Proof. The function $x \mapsto \arcsin^2 x$ is strictly convex for $0 < x < 1$; since a and b are nonconstant affine functions, (10) is strictly convex. Its derivative is negative at $q = 0$ and positive at $q = 1/3$ by (11), so it has exactly one zero between them. The displayed values are reproducible numerical evaluations of that one-dimensional equation, not ingredients of Theorem 3. $\square$

5 Verification, limitations, and novelty check

The script `code/verify_wedge_energy.py` checks the exact orthonormal and regular-simplex values for ambient dimensions 2, 3, 4, tests 60,000 seeded random configurations against (3), and verifies the tight benchmark, positive descent derivative, and scalar cyclic optimum from Theorem 3. The separate exploratory script `code/explore_no_etf.py` performs 250 seeded unconstrained optimizations for $(N, m) = (5, 3)$; every run converged to the cyclic value in Corollary 4. These are sanity checks, not parts of the proof. Run the verifier with

```
conda run --no-capture-output -n sandbox python code/verify_wedge_energy.py
```

The packet still does not characterize the actual global minimizers for $(N, m) = (5, 3)$, nor minimizers for general $N > m + 1$ when no real ETF exists; these are the remaining parts of the source’s broad question. Theorem 3 only rules out all tight frames globally and solves the cyclic subproblem exactly. General frame-potential literature, including [2, 3], motivates tight and equiangular structures, while the tight-frame comparison uses the external universal-optimality theorem of [4].

The run indexes and targeted arXiv/web searches through 19 July 2026 used arXiv:1511.08452, the exact title, and combinations of “wedge energy”, “arcsin squared frame potential”, “Stolarsky”, “equiangular tight frame”, “five points real projective plane”, and “pentagonal frame energy”. They found the source, general frame-potential/ETF papers, and the Cohn–Kumar universal-optimality theorem, but no explicit application to this wedge energy or the non-tightness result above. Novelty confidence remains modest because both arguments combine standard frame and energy bounds.

Human-review recommendation: likely valid partial result. The main checks are strict convexity of Φ , the equality conditions in (5) and Jensen’s inequality, the normalization in the Naimark complement, applicability of [4] to the doubled-angle potential, and the strict sign in (11).

References

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- [2] J. J. Benedetto and M. Fickus, *Finite normalized tight frames*, Advances in Computational Mathematics 18 (2003), 357–385.
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- [4] H. Cohn and A. Kumar, *Universally optimal distribution of points on spheres*, Journal of the American Mathematical Society 20 (2007), 99–148; arXiv:math/0607446.